

EDRF release from canine coronary artery by lectins

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- 1 The effect of wheat germ agglutinin (WGA), concanavalin A (Con A) and lentil lectin agglutinin (LCA) were investigated on pre-contracted canine coronary artery rings *in vitro*.
- 2 In endothelium-intact canine coronary artery, contracted with the thromboxane A₂-analogue, U46619, WGA relaxed the tissue in a concentration-dependent manner, with an inhibitory concentration (IC₅₀) of 112 ± 17 nM ($n = 6$). In the absence of an endothelium, WGA did not cause any relaxation of the tissue.
- 3 In endothelium-intact canine coronary artery, contracted with the thromboxane A₂-analogue, U46619, LCA relaxed the tissue in a concentration-dependent manner, with an inhibitory concentration (IC₅₀) of 423.1 ± 41 nM ($n = 6$). In the absence of an endothelium, LCA produced a $20.1 \pm 1.1\%$ ($n = 6$) relaxation at the highest concentration tested (3 μ M).
- 4 Concanavalin A (Con A) relaxed canine coronary artery in a partial endothelium-dependent manner with an IC₅₀ of 104 ± 19 nM on endothelium-intact coronary artery and an IC₅₀ of 1.3 ± 0.3 μ M ($n = 6$) on endothelium-denuded tissues.
- 5 The relaxation effects of WGA were attenuated by 1 mM N^G-monomethyl L-arginine (L-NMMA) and completely inhibited by haemoglobin (3 μ M), methylene blue (10 μ M) and LY 83583 (30 μ M). Ibuprofen had no effect on WGA-induced relaxation.
- 6 The relaxant effects of WGA were reversed by addition of 20 mM N-acetyl-D-glucosamine (GlcNAc) and N-acetyl-D-galactosamine (GalNAc) but not by α -mannose, D-(+)-galactose, and β -lactose, whereas the endothelin-dependent relaxations to LCA and Con A were unaffected.
- 7 The endothelium-dependent relaxation induced by the lectins was unaffected by pretreatment of the tissue with 1 μ M atropine.
- 8 In the absence of extracellular calcium, WGA was also able to release EDRF suggesting that WGA acts through a second messenger system to release intracellular calcium.
- 9 We suggest that WGA acts as an agonist to release EDRF from endothelial cells possibly by binding to a sugar moiety, specific receptor or adhesion molecules on the endothelial cell surface.

Keywords: EDRF; nitric oxide; endothelium; lectins; wheat germ agglutinin; calcium; canine coronary artery

Introduction

Lectins have been used to characterize and label the vascular endothelial cells without labelling the underlying smooth muscle cells (Raub *et al.*, 1990). Lectins have the ability to bind to saccharide groups of glycoproteins with this binding being reversed by specific sugars (Grulich-Henn *et al.*, 1988) and have recently been shown to be important in mediating cell to cell interactions by regulating biological processes in living organisms (Sharon & Lis, 1989). The characterization of cell surface glycoproteins including growth factor receptors (Sinnott-Smith *et al.*, 1988), and cell adhesion molecules on endothelial cells (Walz *et al.*, 1990) have also used the unique properties of different lectins.

Endothelium-derived relaxing factor (EDRF), a very labile compound with a rapid biological action is released by the vascular endothelium in response to some vasodilators (Furchgott, 1984). EDRF has been shown to be indistinguishable from nitric oxide (NO) with respect to biological activity (Palmer *et al.*, 1987). Vascular responses to EDRF are similar to the responses of nitrovasodilators in that both stimulate guanylate cyclase, which, by increasing smooth muscle guanosine 3':5'-cyclic monophosphate (cyclic GMP) levels, causes smooth muscle relaxation (Furchgott, 1984). This endothelium-dependent vascular relaxation is effectively blocked by haemoglobin, methylene blue, (Martin *et al.*, 1985), N^G-monomethyl-L-arginine (L-NMMA) (Rees *et al.*, 1989) and LY 83583 (6-anilino-5,8-quinolinedione) (Mulsch

et al., 1988). In cultured endothelial cells, the stimulation of the same receptors causes a coupled release of EDRF and prostacyclin (PGI₂) which results in vasodilatation (De Nucci *et al.*, 1988). Thrombin stimulates prostacyclin (PGI₂) synthesis in human umbilical vein endothelial cells (HUVEC) (Weksler *et al.*, 1978) and it has been shown that wheat germ agglutinin (WGA) blocks thrombin-induced stimulation of [Ca²⁺]_i, [³H]-arachidonate release, and prostacyclin production in cultured HUVECs (Grulich-Henn *et al.*, 1988). We have suggested that, in the rabbit aorta, WGA binds to a receptor or a cell surface glycoprotein which, via the release of EDRF from the endothelial cell, causes smooth muscle relaxation (Kleha *et al.*, 1991). The purpose of the present investigation was to determine, in more detail, the mechanism(s) involved in the release of EDRF by WGA (Kleha *et al.*, 1991). Lentil lectin agglutinin (LCA) and concanavalin A (ConA) were also investigated to determine if other lectins, with different sugar specificities, caused similar effects to wheat germ agglutinin on vascular tissue.

Methods

Isometric muscle tension bath system

Mongrel dogs were anaesthetized with sodium pentobarbitone (35 mg kg⁻¹) by intravenous injection into the saphenous vein. The heart and carotid arteries were excised and placed in cooled HEPES buffered physiological salt solution (PSS). The PSS had the following composition (mM): NaCl 140, KCl 5, CaCl₂ 1.5, MgCl₂ 1, D-glucose 10, N-[2-

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hydroxyethyl] piperazine-N'-[2-ethanesulphonic acid] (HEPES) 2.7, sodium N-[2-hydroxyethyl] piperazine-N'-[2-thanesulphonic acid] (Na HEPES) 2.3, ethylenediamine-tetraacetic acid (EDTA) 10 μ M and ibuprofen 10 μ M. This solution had a pH 7.4 at 37°C when bubbled with air and is identical to that used previously to determine the pharmacological, biochemical and electrophysiological characteristics of cultured endothelial cells (Johns *et al.*, 1987; Freay *et al.*, 1989; Laskey *et al.*, 1990). The circumflex branch of the coronary artery was isolated and any traces of adhering fascia removed by careful dissection. Rings of approximately 3 mm in length were used with the endothelium layer intact or the endothelium was removed by gently rubbing the intimal surface of the blood vessel with a wooden toothpick.

Tissues were suspended in 5 ml isometric muscle tension baths under 4 g of tension in PSS bubbled with room air at 37°C. The tension was measured with a force transducer (Grass FT03C) and displayed on a Beckman polygraph (R-611). The vessels were allowed to equilibrate for 2 h, during which time the maximum contractile response was measured by the addition of 40 mM KCl every 30 min. A constant basal tension of 4 g was maintained during this period. The presence of an endothelial cell layer was confirmed by the addition of 300 nM acetylcholine (ACh), which is known to cause a maximum relaxation (DeMey & Vanhoutte, 1981), following development of a maintained contraction to added KCl (20 mM).

In order to determine the effects of the different lectins the canine coronary artery was contracted with 50 nM of the thromboxane A₂-analogue, U46619. After a sustained contraction developed, concentration-response curves to the lectins were constructed.

Calcium-free (EGTA) experiments

Endothelium-intact and endothelium-denuded canine coronary arteries were prepared as described above. After equilibration, the PSS was replaced with calcium-free PSS containing 2 mM EGTA. The calcium-free PSS (2 mM EGTA) solution was prepared by substitution of calcium chloride with 2 mM of the sodium salt of ethylene glycol-bis (β -aminoethyl ether) N,N,N',N'-tetraacetic acid (EGTA). After incubation in this solution for 2 min the tissues were contracted by addition of endothelin-1 (final concentration 30 nM). After a further 7 min, WGA was added to the baths (final concentration 900 nM). After the solution bathing the tissues had been replaced with the calcium-free (2 mM EGTA) PSS the tissues were not again exposed to calcium containing PSS. Data are expressed as percentage relaxation of endothelin-1 contraction and shown as mean $\pm$ s.e.mean.

Superfusion bioassay

The bioassay tissue (endothelium-denuded canine coronary artery) was mounted between two stainless steel hooks under a resting tension of 3 g. The tissue was superfused with prewarmed, aerated (37°C) PSS at a rate of 4 ml min⁻¹. The tissue was allowed to equilibrate for 2 h during which time responses to addition of 40 mM KCl to the PSS were determined. The absence of a functional endothelium on the canine coronary artery was confirmed by the lack of a relaxation of the bioassay tissue to acetylcholine (300 nM) following a 20 mM KCl-induced contraction.

Canine carotid arteries were mounted on stainless steel cannulae in a jacketed organ bath. The carotid artery (donor tissue) was perfused with PSS at a rate of 4 ml min⁻¹ at 37°C. The bioassay tissue was contracted by changing the solution perfusing the donor to one containing 10 nM U46619. Basal release of EDRF was measured by moving the canine carotid artery (donor) over the canine coronary artery and allowing the perfusate to superfuse the bioassay tissue. Stimulated release of EDRF was measured by perfusing the carotid artery with a solution containing 900 nM WGA and

10 nM U46619 in PSS and allowing the perfusate to superfuse the bioassay tissue. Data are expressed as percentage relaxation of thromboxane A₂-analogue U46619 and shown as mean $\pm$ s.e.mean.

Drugs

All the chemicals used were of analytical grade and obtained from Sigma Chemical Co., St Louis, MO., U.S.A. The following drugs were used: N-acetyl-D-galactosamine (GalNAc, Sigma Chemical Co., St Louis, MO., U.S.A.), N-acetyl-D-glucosamine (GlcNAc, Sigma Chemical Co., St Louis, MO., U.S.A.), acetylcholine chloride (ACh, Sigma Chemical Co., St Louis, MO., U.S.A.), 6-anilino-5,8-quinolinedione (LY 83583, Lilly, U.S.A.), atropine sulphate (ICN, U.S.A.), concanavalin A lectin (Con A, Boehringer Mannheim Biochemicals, U.S.A.), 9,11-dideoxy-11 α ,9 α -epoxy-methanoprostaglandin F_{2 α} (U46619, Sigma Chemical Co., St Louis, MO., U.S.A.), endothelin (ET-1, Mid-West Biotech, U.S.A.), D-(+)-galactose (Sigma Chemical Co., St Louis, MO., U.S.A.), haemoglobin (prepared as described by Martin *et al.*, 1985) (Hb, Sigma Chemical Co., St Louis, MO., U.S.A.), ibuprofen, free acid (Ibu, Sigma Chemical Co., St Louis, MO., U.S.A.), β -lactose (Sigma Chemical Co., St Louis, MO., U.S.A.), *Lens culinaris* (Lentil) agglutinin lectin (LCA, Boehringer Mannheim Biochemicals, U.S.A.), α -mannose (Sigma Chemical Co., St Louis, MO., U.S.A.), N^G monomethyl-L-arginine acetate (L-NMMA, Sigma Chemical Co., St Louis, MO., U.S.A.), methylene blue (Mb, Sigma Chemical Co., St Louis, MO., U.S.A.), wheat germ agglutinin lectin (WGA, Boehringer Mannheim Biochemicals, U.S.A.).

Statistical analysis

Results are expressed as mean $\pm$ standard error of the mean (s.e.mean). Statistical significance between data points was tested by Student's unpaired *t* test. Probability of $P < 0.05$ was considered significant and indicated as *.

Results

The effects of WGA on endothelium-intact and endothelium-denuded canine coronary arteries are shown in Figure 1. Representative tension recordings of paired canine coronary arteries are shown in Figure 1a. WGA was added to the tissue preparations in a concentration-dependent manner from 30 nM up to 900 nM as described in Methods. WGA caused endothelium-dependent relaxations with a half maximal inhibitory concentration (IC₅₀) of 112 $\pm$ 17 nM ($n = 6$). No vasorelaxation occurred in the endothelium-denuded preparations as shown in Figure 1b ($n = 6$).

Concentration-dependent responses to the lectin LCA on endothelium-intact and endothelium-denuded preparations of canine coronary artery are shown in Figure 2. In six experiments, LCA caused the endothelium-intact tissues to relax with the maximum concentration (3 μ M) yielding a 101.2 $\pm$ 3.5% relaxation and an IC₅₀ of 423.1 $\pm$ 41 nM. LCA caused the endothelium-denuded tissue to relax 20.1 $\pm$ 1.1% at the maximum concentration tested (3 μ M) but did not reach an IC₅₀.

The lectin, Con A, caused both endothelium-intact and endothelium-denuded canine coronary artery preparations to relax, however the extent of vasodilatation was more pronounced in the endothelium-intact preparations. Concentration-dependent responses to Con A on canine coronary arteries with and without endothelium are shown in Figure 3. At the maximum concentration tested (1.4 μ M) on endothelium-intact preparations, Con A gave a relaxation response of 102 $\pm$ 3.3% with an IC₅₀ of 104 $\pm$ 19 nM as compared to a 61.8 $\pm$ 7.9% on endothelium-denuded tissues ($n = 6$).

Figure 4 demonstrates the stimulated release of EDRF

from the endothelium induced by WGA. The effects of WGA were specifically inhibited by haemoglobin, methylene blue, and LY 83583. The recovery response of 10 μ M haemoglobin on WGA-induced relaxations is shown in Figure 4a. In 6

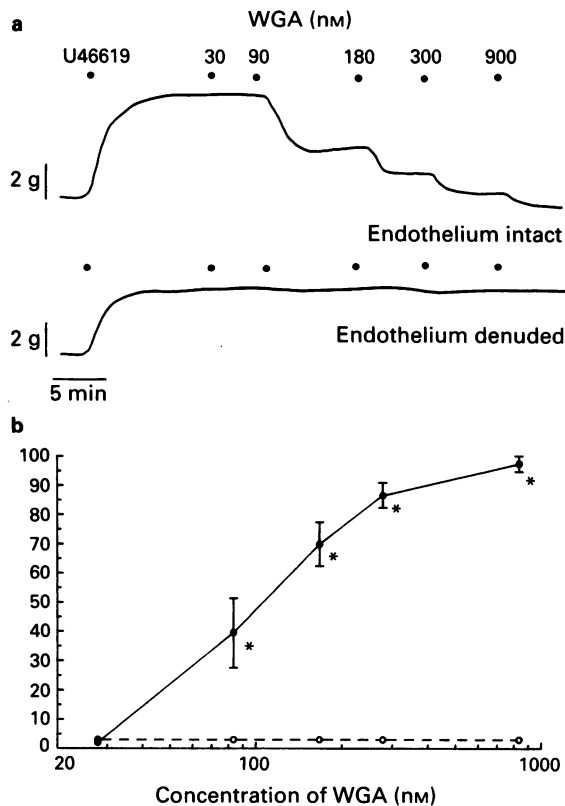


Figure 1 (a) The effects of wheat germ agglutinin (WGA) on canine coronary artery. Endothelium-intact and denuded rings of canine coronary arteries were contracted with 50 nM U46619. Concentration-responses to WGA were performed in the following sequence: 30 nM, 90 nM, 180 nM, 300 nM, 900 nM. Representative traces of coronary artery rings showing endothelium-dependent relaxations to wheat germ agglutinin (WGA) in a concentration-dependent manner. (b) The effects of different concentrations of WGA on rings of canine coronary artery in the presence (●) and absence (○) of an endothelium. The values shown are mean $\pm$ s.e. mean relaxations of a 50 nM U46619 contraction (6 experiments). All relaxations induced by concentrations of WGA above 30 nM were significantly ($P < 0.05$) different and indicated as (*).

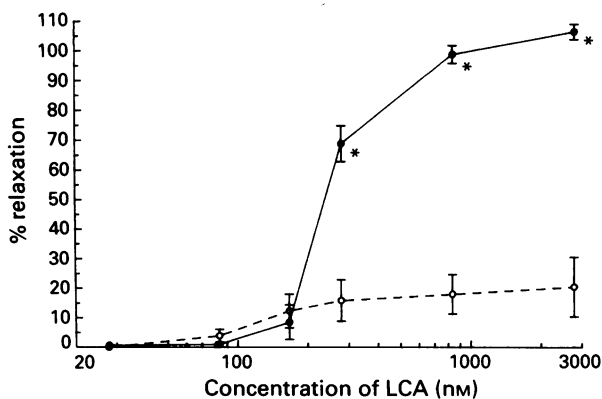


Figure 2 The effects of different concentrations of lentil (*Lens culinaris*) agglutinin (LCA) on coronary artery rings in the presence (●) and absence (○) of an endothelium. The values shown are mean $\pm$ s.e. mean relaxations of a 50 nM U46619-induced contraction (6 experiments). Significantly ($P < 0.05$) different points are indicated (*).

experiments, haemoglobin caused a $114.9 \pm 6.5\%$ recovery when compared to the U46619 (50 nM)-induced contraction. Methylene blue at 3 μ M reversed the WGA-induced relaxations by $89.7 \pm 10.5\%$ ($n = 6$), with a representative tracing shown in Figure 4b. Figure 4c shows a representative trace of the effects of the guanylate cyclase inhibitor, LY 83583, at a concentration of 30 μ M. LY 83583 produced a $94.9 \pm 6.7\%$ reversal of the WGA-induced relaxation ($n = 6$) of paired canine coronary arteries. WGA binds specifically to N-acetyl-D-glucosamine moieties with a high affinity. To show the effects of WGA were due to the lectin effect and not a possible contaminant in the solution, N-acetyl-D-glucosamine was added to the bath after WGA-induced relaxation. The specific effect of GlcNAc on relaxations induced by WGA is

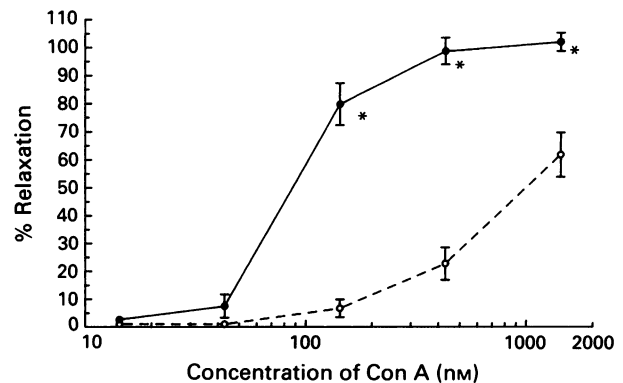


Figure 3 The effects of different concentrations of concanavalin (Con A) on canine coronary artery rings in the presence (●) and absence (○) of endothelium. The values shown are mean $\pm$ s.e. mean relaxations of a 50 nM U46619-induced contraction (6 experiments). Significantly ($P < 0.05$) different points are indicated (*).

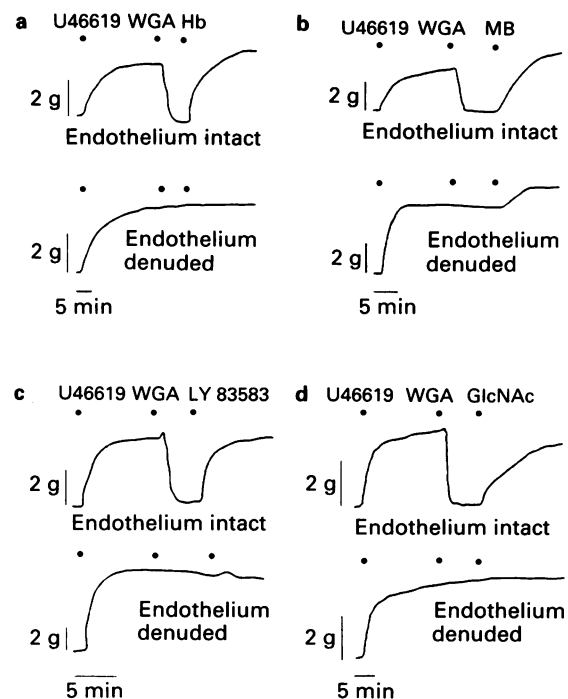


Figure 4 The effects of haemoglobin (Hb), methylene blue (MB), LY 83583, and N-acetyl-D-glucosamine (GlcNAc) on relaxations induced by wheat germ agglutinin (WGA) of the canine coronary artery. Endothelium-intact and denuded coronary arteries were contracted with 50 nM U46619, and the response to 900 nM WGA was measured. Representative isometric tension recordings of the inhibitory effects of 10 μ M bovine haemoglobin, 3 μ M methylene blue and 3 μ M LY 83583 are shown in (a), (b) and (c) respectively. The effects of GlcNAc on the WGA-induced relaxation is shown in (d).

shown in Figure 4d. The sugar at 20 mM reversed the relaxation by $98.7 \pm 2.8\%$ ($n = 6$).

The EDRF/NO synthase inhibitor L-NMMA (1 mM) was preincubated with endothelium-intact canine coronary arteries for 5 min before responses to increasing concentrations of WGA were measured. Figure 5 shows that 1 mM L-NMMA decreased the sensitivity of the tissue to relaxations induced by WGA. The IC_{50} for the concentration-response curve to WGA in the presence of 1 mM L-NMMA was 341 ± 115 nM and is shown in Figure 5a. In the absence of L-NMMA, the IC_{50} for WGA was 112 ± 17 nM. There was a significant shift in the concentration-response curve in the presence of 1 mM L-NMMA illustrating that L-NMMA, by inhibiting nitric oxide synthase (known to be important for EDRF synthesis) is attenuating the WGA-induced relaxation (Figure 5a). Figure 5b demonstrates the effects of L-NMMA on responses of the canine coronary artery to acetylcholine. These experiments illustrate that the acetylcholine-induced relaxation of the dog coronary artery is also significantly attenuated by 1 mM L-NMMA.

The effects of WGA on canine carotid arteries in the cascade superfusion bioassay system using endothelium-denuded canine coronary artery as the bioassay tissue and

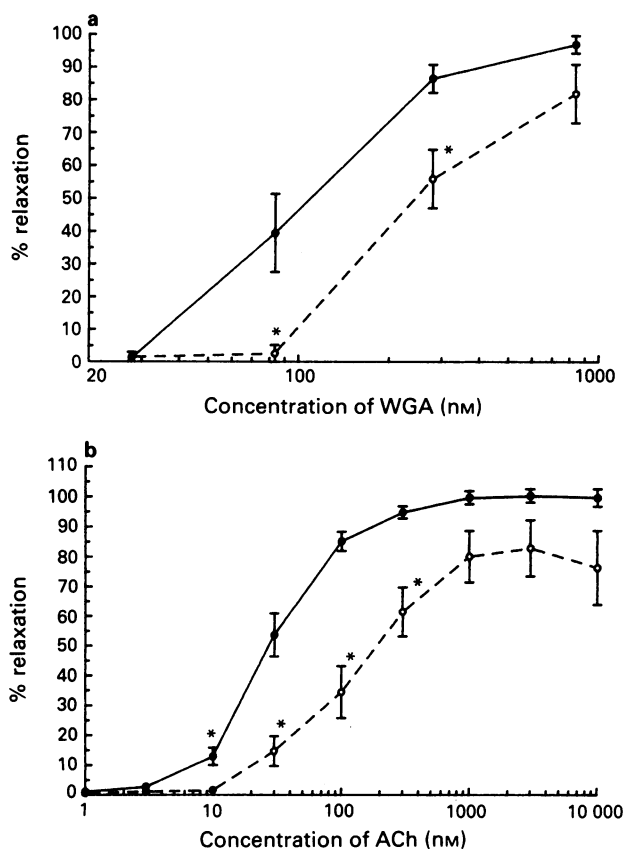


Figure 5 (a) The effects of N^G-monomethyl-L-arginine (L-NMMA) on the response of canine coronary artery to wheat germ agglutinin (WGA). Endothelium-intact canine coronary arteries were pretreated with 1 mM L-NMMA for 5 min before a contraction to 50 nM U46619 was measured, after which a concentration-response curve to WGA was obtained. Curves shown are endothelium-intact coronary artery in the presence (○) and absence (●) of 1 mM L-NMMA. Data are expressed as mean $\pm$ s.e.mean relaxations, (6 experiments). (b) The effects of L-NMMA (1 mM) on the response of the canine coronary artery to acetylcholine (ACh). Endothelium-intact canine coronary arteries were pretreated with 1 mM L-NMMA for 5 min before a concentration-response curve to acetylcholine was obtained. Curves shown are endothelium-intact coronary artery in the presence (○) and absence (●) of 1 mM L-NMMA. Data are expressed as mean $\pm$ s.e.mean relaxations (6 experiments). Significantly ($P < 0.05$) different points are indicated (*).

canine carotid artery as the donor were investigated. Endothelium-denuded canine coronary artery was contracted with 10 nM U46619. After a sustained contraction was measured, basal EDRF was then measured after which the canine carotid artery was perfused with 900 nM WGA and a stimulated relaxation was measured. A representative trace is shown in Figure 6a. Comparison of basal EDRF release from canine carotid arteries with WGA stimulated EDRF release plus basal EDRF release is shown in Figure 6b. In 6 experiments, basal EDRF relaxations from carotid arteries measured $51.9 \pm 8.7\%$ when compared to the 10 nM U46619-induced contraction, while the WGA-stimulated release of EDRF plus basal EDRF release was measured at $82.1 \pm 4.4\%$ resulting in an overall WGA stimulation of relaxation of $28.6 \pm 5.5\%$ as compared to a 10 nM U46619 contraction. This further demonstrates the stimulated release of EDRF by WGA.

Atropine effectively inhibits vasodilatation induced by acetylcholine in rabbit aorta precontracted with noradrenaline (Furchgott & Zawadzki, 1980). The effects of atropine on the response of canine coronary arteries to both acetylcholine and WGA were investigated. Endothelium-intact and denuded coronary arteries were contracted with 50 nM of the thromboxane A₂-analogue, U46619. After a sustained contraction was obtained, 1 μ M atropine was incubated with the tissue for 5 min after which 300 nM acetylcholine was added. After blockade of the acetylcholine response by 1 μ M atropine, a response to 900 nM WGA was measured. There was no relaxation response to 300 nM acetylcholine in the presence of 1 μ M atropine ($0 \pm 0\%$, $n = 6$). The vasodilator response to 900 nM WGA in the presence of 1 μ M atropine plus 300 nM acetylcholine was $97.6 \pm 2.4\%$ ($n = 6$) demonstrating that WGA does not cause vasodilatation by an effect

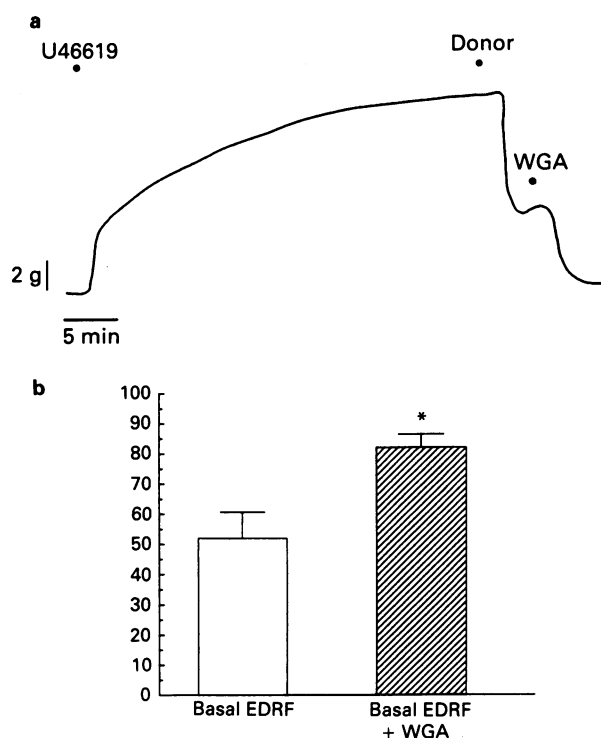


Figure 6 The stimulated release of EDRF from perfused canine carotid arteries by wheat germ agglutinin (WGA). Representative tracing of an endothelium-denuded canine coronary artery (bioassay tissue) superfused with effluent from a perfused canine carotid artery showing basal EDRF and WGA stimulated EDRF is shown in (a). Graph comparing basal release of EDRF with stimulated release by 900 nM WGA is shown in (b). The values shown are mean (with s.e.mean) relaxations of tissues contracted with 10 nM U46619 (6 experiments). Significantly ($P < 0.05$) different points are indicated (*).

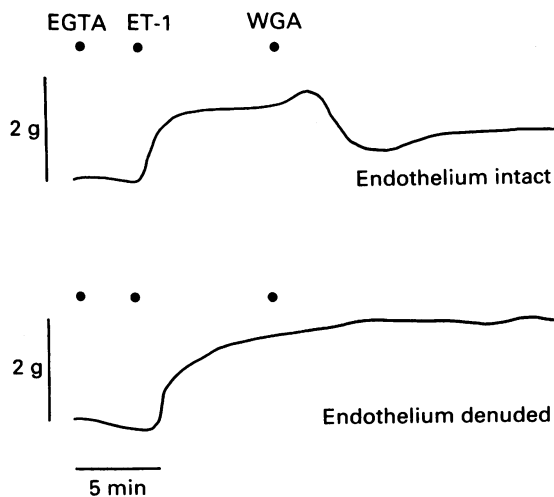


Figure 7 The relaxation effects of wheat germ agglutinin (WGA) on canine coronary artery in the absence of extracellular calcium. Endothelium-intact and endothelium-denuded arteries were contracted with 30 nM endothelin-1 in calcium-free, 2 mM EGTA PSS. A significant relaxation to 900 nM WGA was observed in endothelium-intact tissues with a representative tracing shown in (a). Only a slight relaxation was measured in endothelium-denuded tissues and is shown in (b).

on muscarinic acetylcholine receptors to stimulate EDRF release.

In order to test the possibility that WGA was acting on the thrombin receptor a series of 5 experiments was performed on tissues in which tachyphylaxis to thrombin had been induced by repetitive exposure. In tissues contracted with U46619, thrombin (2 iu ml^{-1}) was added three times which caused $100.9 \pm 4.0\%$, $75.1 \pm 10.0\%$ and $45.2 \pm 5.8\%$ relaxations respectively. In matched tissues the first two relaxations to thrombin were not significantly different from the control tissue ($105.4 \pm 3.2\%$ and $64.4 \pm 14.3\%$) whereas the third response to 900 nM WGA ($117.1 \pm 18.4\%$) was significantly different from the thrombin response ($P > 0.05$).

Representative traces of the effects of WGA on canine coronary arteries with the endothelium-intact and endothelium-denuded in 2 mM EGTA calcium-free PSS are shown in Figure 7. In 5 experiments, 900 nM WGA caused an $80.6 \pm 3.4\%$ relaxation in endothelium-intact canine coronary arteries and a $17.2 \pm 5.5\%$ relaxation in endothelium-denuded arteries. In comparison, 900 nM WGA relaxed endothelium-intact canine coronary arteries $98.9 \pm 2.7\%$ in PSS containing 1.5 mM calcium chloride but there were no relaxations measured in endothelium-denuded canine coronary arteries.

Discussion

Wheat germ agglutinin is a pure protein with a molecular weight of 36,000 and is composed of two similar peptide chains with two polypeptide chains that contain sugar binding sites (Nagata & Burger, 1974). Most lectins have been used to characterize cell types and have been used extensively to isolate and partially purify many proteins including receptors for hormones, neurotransmitters, growth factors and immunoglobins (Hedo, 1984) along with adhesion molecules such as the endothelial leukocyte adhesion molecule-1 (ELAM-1) (Walz *et al.*, 1990). We have previously shown that WGA stimulates the release of EDRF from rabbit aorta in a concentration-dependent manner (Kleha *et al.*, 1991).

The endothelium-dependent vasodilatation induced by WGA in canine coronary artery was inhibited by haemoglobin, methylene blue, LY 83583, and L-NMMA, all of

which are known to inhibit pathways that lead to EDRF/NO-induced relaxation (Martin *et al.*, 1985; Mulisch *et al.*, 1988; Rees *et al.*, 1989). The endothelium-dependent relaxations were observed in the presence of ibuprofen suggesting that the relaxations are not due to the release of prostacyclin. The relaxations to WGA were specifically reversed by the sugar N-acetyl-D-glucosamine; however, this sugar had no effect on the partial endothelium-dependent relaxation of the dog coronary artery to Con A. The vasodilator effect of WGA was shown to be unaffected by atropine (known to inhibit acetylcholine-induced relaxation of rabbit aorta; Furchgott & Zawadzki, 1980) suggesting that WGA was not working on the ACh muscarinic receptor and probably causing the vasodilator response by acting on its own specific receptor.

The possibility that WGA may act as an agonist on the thrombin receptor was also tested. It was demonstrated that WGA was still able to produce full relaxation of tissues when the vasodilator response to thrombin had been reduced substantially by repeated exposure to thrombin.

The endothelium-dependent relaxations induced by WGA and LCA along with the partial endothelium-dependent relaxations induced by Con A on canine coronary artery demonstrate that there might be different lectin binding molecules present on the endothelial cell surface. However, it is possible that this result might be due to the different glycosylation states of the same receptor recognizing the different lectins on the endothelial cell surface.

In the absence of extracellular calcium and in the presence of 2 mM EGTA, endothelin has been shown to induce a maintained contraction (Pang *et al.*, 1990). This calcium-free contraction was utilized in order to determine if, in uncultured native endothelium, WGA could induce EDRF release. In cultured endothelial cells it has been demonstrated that EDRF release can occur in the absence of extracellular calcium via activation of phospholipase C (Freay *et al.*, 1989) and release of intracellular calcium. In calcium-free solutions we have demonstrated that WGA could release EDRF suggesting that it is working via a receptor-linked system rather than via a non-specific increase in cell permeability to extracellular calcium.

EDRF is released by endothelial cells and induces vasodilatation by activating guanylate cyclase in the underlying vascular smooth muscle (Ignarro *et al.*, 1986; 1987; Angus & Cocks, 1989; Murad & Ishii, 1990). Aggregating platelets cause a transient increase in tension on endothelium-denuded canine coronary arteries and a profound relaxation of endothelium-intact arteries (Cohen *et al.*, 1983; Houston *et al.*, 1986). Although this effect was attributed to factors released from the platelets, a direct interaction of the platelets with the endothelium was not ruled out and a direct interaction of blood elements, including platelets, with the endothelium has recently been described (Flier & Heistad, 1993). The effects of WGA on canine coronary artery endothelium would suggest a possible lectin recognition molecule on the surface of endothelial cells. The existence of lectin receptors linked to tyrosine kinase have been reported in human platelets (Inazu *et al.*, 1991) and a receptor linked to mobilization of intracellular calcium has been reported in human lymphoblastoma cells (Tordai *et al.*, 1989; Tregear *et al.*, 1991) and rat lymph node cells (Tregear *et al.*, 1991). The ability of WGA to aggregate human platelets is well documented with the capacity of this binding being dependent on the presence of platelet surface sialic acid (Ganguly & Fossett, 1984) and may involve the activation of a tyrosine kinase signal transduction system (Inazu *et al.*, 1991). Therefore, the lectins WGA, Con A and LCA could recognize a receptor (or receptors) on the endothelial cell surface which stimulate EDRF release by mechanisms already described in other cell types (Tordai *et al.*, 1989; Inazu *et al.*, 1991) and this receptor could be an adhesion molecule (or molecules) that has the ability to recognize different cell-specific, lectin-type molecules. Whether 'lectin receptors' are

linked to, or indistinguishable from, any of the many adhesion molecules has yet to be determined.

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